THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 12 & 13: **How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** abiotic components	Air, water, soil, temperature
2	**(c)** aquatic ecosystem	Water-based
3	**(d)** carnivores	Flesh-eaters
4	**(d)** recycling nutrients	Nutrients return to soil
5	**(b)** frog	Eats a herbivore
6	**(c)** Goldilocks zone	Neither too hot nor too cold
7	**(d)** harmful ultraviolet rays	Absorbs UV
8	**(a)** a strong greenhouse effect	Dense CO_2 atmosphere
9	**(a)** life	All living things
10	**(b)** two parents and fusion of gametes	Variation arises

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One benefits, the other is unaffected (1). Orchid on a tree (1).	1+1
12	One organism benefits and the other is harmed (1). Tapeworm in human gut (1).	1+1
13	Ensures continuity of life (1). Without it species would disappear (1).	1+1
14	Asexual: single parent, identical offspring (1). Sexual: two parents, variation in offspring (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Deforestation (1). Pollution (1). Invasive species or overexploitation (1).	1+1+1
16	Offspring show new features (1½). Variation helps adapt to changing environments (1½).	1½+1½
17	Protect habitats (1½). Conserve biodiversity (1½).	1½+1½
	OR: Bandipur – Karnataka (1). Manas – Assam (1). Chilika – Odisha (1).	1+1+1
18	Burning fossil fuels (1). Deforestation (1). Water and air pollution (1).	1+1+1
	OR: Delicate balance of climate (1). Changes affect weather and ecosystems (1). Could endanger species (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Deforestation (1). Pollution (1). Climate change and invasive species (1). Plant trees (1). Reduce, reuse, recycle (1).	1+1+1+1+1
	OR: National parks (1). Sanctuaries (1). Biosphere reserves (1). Community areas (1). Importance (1).	1+1+1+1+1
20	Soil gives nutrients (1). Rocks and minerals provide materials (1). Plants make food (1). Animals consume and pollinate (1). Microbes recycle nutrients (1).	1+1+1+1+1
	OR: Species depend on each other (1). Changes affect food chains (1). Example of predator-prey (1). Balanced temperature and oxygen (1). Disturbance leads to crisis (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 12 & 13: **How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** decrease	Less food available
2	**(a)** parasitism	Parasite benefits, host harmed
3	**(d)** organic manure	Vrikshayurveda advocates it
4	**(d)** 1987	As per the chapter
5	**(a)** Karnataka	Where predators and prey were studied
6	**(c)** seed	Inside the fruit
7	**(d)** inside the female body	Internal fertilisation
8	**(d)** biodiversity loss	Three linked threats
9	**(b)** Reuse, repair and recycle	Reduce pollution
10	**(b)** endanger life on Earth	Delicate balance

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Clean air and water (1). Food, medicine or climate regulation (1).	1+1
12	Deforestation (1). Pollution or overexploitation (1).	1+1
13	Pollination: transfer of pollen to stigma (1). Fertilisation: fusion of male and female gametes (1).	1+1
14	Cell formed after fusion of gametes (1). It develops into an embryo (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Provide conditions to live (1½). Examples: sunlight, water, soil, temperature (1½).	1½+1½
16	More UV rays reach Earth (1½). They damage cells and ecosystems (1½).	1½+1½
17	Natural: forest, pond (1½). Human-made: farm, fish pond (1½).	1½+1½
	OR: Energy is lost at each level (1½). Less energy supports fewer organisms (1½).	1½+1½
18	Decompose waste (1). Fix nitrogen (1). Help in cycles of nutrients (1).	1+1+1
	OR: Sustainable use of resources (1). Waste management (1). Planting and protecting trees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Algae → small fish → big fish (1). (ii) Producer algae, consumers fish (1). (iii) Decomposers recycle (1). (iv) Big fish suffer (1). Algae increase (1).	1+1+1+1+1
	OR: Water decreases (1). Plants dry (1). Herbivores starve (1). Carnivores decline (1). Decomposers also affected (1).	1+1+1+1+1
20	Not too hot, not too cold (1). Liquid water (1). Atmosphere (1). Gravity (1). Protective fields (1).	1+1+1+1+1
	OR: Number of parents (1). Gametes (1). Variation (1). Speed (1). Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 12 & 13: **How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** soil	Non-living
2	**(b)** terrestrial ecosystem	Land-based
3	**(c)** omnivores	e.g. humans, crows
4	**(d)** food chain	Linear feeding order
5	**(b)** eagle	Fourth trophic level
6	**(c)** temperature is not too extreme during the year	Constant distance from Sun
7	**(c)** three oxygen atoms	O_3
8	**(c)** air	Mixture of gases
9	**(c)** interact and sustain life	Interconnected
10	**(c)** fertilisation	Forms zygote

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Non-native species that spread widely (1). They harm native species (1).	1+1
12	Growing the same crop repeatedly (1). Reduces biodiversity and affects pollinators (1).	1+1
13	Sugarcane (1). Rose or potato (1).	1+1
14	Climate change (1). Pollution and biodiversity loss (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dead matter would pile up (1). Nutrients not recycled (1). Plants and ecosystem would suffer (1).	1+1+1
16	Many species perform different roles (1½). If one is lost, others keep balance (1½).	1½+1½
17	They make food using sunlight (1½). All consumers depend directly or indirectly on them (1½).	1½+1½
	OR: Clean air (1). Fresh water (1). Food and medicine (1).	1+1+1
18	Plant greenhouse traps warm air in closed glass space (1½). Greenhouse effect traps heat by gases in atmosphere (1½).	1½+1½
	OR: Air provides oxygen for respiration (1). Water is needed by all organisms (1). Sunlight supports photosynthesis (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Clean air and water (1). Food and medicine (1). Climate regulation (1). Human survival depends (1). Conservation protects future (1).	1+1+1+1+1
	OR: Vrikshayurveda text (1). Organic manure (1). Kunapa Jala (1). Chemical farming harms soil (1). Balanced approach (1).	1+1+1+1+1
20	Temperature rises (1). Weather changes (1). Sea levels change (1). Save energy (1). Plant trees and use clean energy (1).	1+1+1+1+1
	OR: Air pollution (1). Water pollution (1). Soil pollution (1). Waste reduction (1). Recycling (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 12 & 13: **How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** both organisms benefit	e.g. flower and bee
2	**(d)** clean air, water, food and climate regulation	Benefits from nature
3	**(c)** liquid fertiliser from animal and plant waste	Made by fermentation
4	**(b)** deforestation	Habitat loss
5	**(a)** producer → herbivore → carnivore	Eaten by next level
6	**(a)** stem cuttings	Vegetative propagation
7	**(d)** after the egg is laid	During hatching
8	**(a)** greenhouse gases like carbon dioxide	Raises Earth's temperature
9	**(d)** Using public transport or cycling	Environment friendly
10	**(a)** using natural resources sustainably	Community role

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They slow winds and waves in storms (1). They are home to many species (1).	1+1
12	National parks (1). Wildlife sanctuaries or biosphere reserves (1).	1+1
13	Reduce, reuse and recycle (1). Use clean energy and public transport (1).	1+1
14	To avoid competition for resources (1). To grow in new places (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tigers eat deer and wild boar (1). If prey declines, tigers starve (1). Needs healthy forests (1).	1+1+1
16	No oxygen or ozone shield (1). Temperature would vary greatly (1). Life could not survive (1).	1+1+1
17	They depend on the ecosystem (1½). Their knowledge helps manage it (1½).	1½+1½
	OR: Plant trees (1). Reduce pollution and plastic (1). Save water and energy (1).	1+1+1
18	Reduce waste (1½). Save resources and energy (1½).	1½+1½
	OR: Species disappear due to habitat loss (1½). Ecosystems become weaker (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terrestrial examples (1). Aquatic examples (1). Abiotic differences (1). Organisms (1). Dependence between them (1).	1+1+1+1+1
	OR: Web diagram with arrows (1). Producer identified (1). Consumers levels named (1). Removal increases mice and frogs (1). Hawks lose a food source and balance changes (1).	1+1+1+1+1
20	Save energy (1). Save water (1). Reduce plastic (1). Plant trees (1). Spread awareness (1).	1+1+1+1+1
	OR: Some oxygen converts into ozone, a three-atom form (1). Ozone layer absorbs ultraviolet rays (1). Pollutant particles can reduce it (1). More UV reaches the surface (1). Life and ecosystems get harmed (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.